Or Goldreich

Statistics · Stochastic optimization · Probabilistic modeling

Chicago, Illinois · orgoldreich@gmail.com

Research interests

Convergence, stability and statistical properties of stochastic gradient descent; learning-rate schedules, gradient noise, concentration inequalities and high-dimensional statistical theory.

Education

The University of Chicago September 2022 - present
PhD in Statistics

Tel Aviv University October 2021 - June 2022
Graduate coursework in Statistics and Data Science
Transferred to the University of Chicago PhD program before completing the degree.

Technion - Israel Institute of Technology October 2015 - September 2018
BSc in Mathematics, cum laude
GPA: 91.1. Dean’s List on four occasions.

Papers

Or Goldreich, Ziyang Wei, Soham Bonnerjee, Jiaqi Li, Wei Biao Wu. **Asymptotic theory of SGD with a general learning-rate.** NeurIPS 2025.

Or Goldreich, Soham Bonnerjee, Qianqian Lei, Jiaqi Li, Wei Biao Wu. **Identifying stability regions of SGD with constant learning rates.**

Or Goldreich, Wei Biao Wu. **Self-Normalized Cramér-Type Moderate Deviations for SGD.**

Or Goldreich, Wei Biao Wu. **High-Probability Last-Iterate Bounds for Clipped SGD Under Heavy-Tailed Noise.**

Or Goldreich, Qianqian Lei, Jiaqi Li, Sung-Yi Lin, Wei Biao Wu. **L^p Convergence of Stochastic Gradient Descent with Mixture Learning Rates.**

Engineering and research experience

Nvidia Networking February 2020 - November 2021
Formal Verification Software Engineer · Tel Aviv-Yafo
Developed verification methodology and Python/C tooling for network protocols and low-level dataplane code. Built reusable macros and automation scripts; consulted on numerical logic, correctness and edge cases.

Tel Aviv University February 2022 - August 2022
Research Assistant · Tel Aviv-Yafo
Provided mathematical and quantitative support for empirical legal research in antitrust and regulatory policy. Helped formulate quantitative arguments, structure empirical analyses and interpret model outcomes.

Israeli Air Force January 2012 - January 2015
Computing Systems Instructor · Tzrifin
Managed a logistics information system and delivered technical support and user training. Designed and implemented a C#/.NET/SQL application for time tracking and scheduling.

Teaching

The University of Chicago January 2023 - present
Teaching Assistant
Office hours and grading for STAT 23400 Statistical Models and Methods, STAT 22600 Analysis of Categorical Data, STAT 26100 Time Dependent Data and STAT 22400 Applied Regression Analysis.

Technion - Israel Institute of Technology October 2017 - January 2020
Teaching Assistant
Taught linear and abstract algebra, set theory and mathematical foundations. Prepared original recitation notes; received recognition for excellence in teaching in three consecutive semesters.

Skills, languages and service

Technical: Python, R, SQL, MATLAB, C, C++, C#, Java; .NET application development.

Languages: Hebrew (native); English (fluent).

Service: Technion department student committee member for two years.